

Satilmis Furkan KAHRAMAN

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RESEARCH INTERESTS	Medical Imaging, Image Processing, Signal Processing, Hardware Design and Implementation
EDUCATION	Bilkent University, Ankara, Turkey <i>Master of Science</i>, Electrical and Electronics Engineering (English) August 2025 - June 2028 (Expected) CGPA: 3.42/4.00 Bilkent University, Ankara, Turkey <i>Bachelor of Science</i>, Electrical and Electronics Engineering (English) August 2020 - June 2025 CGPA: 3.38/4.00
PUBLICATION	• S. Furkan Kahraman, Ege Kor, Emine Ulku Saritas, "Single-Sided MPI without Selection Fields: Proof-of-Concept Results," Published in International Journal on Magnetic Particle Imaging (IJMPI), 2025.
HONOR & AWARDS	• Ranked 61st among 2.5 million examinees in the nationwide university entrance examinations, Turkey, 2020. • Comprehensive scholarship covering full tuition and monthly stipend for undergraduate education in the top engineering school in Turkey, Bilkent University, Ankara, Turkey, 2020-2025. • Research Excellence Award by Bilkent Electrical and Electronics Engineering Department, 2025.
ACADEMIC EXPERIENCE	Undergraduate Researcher, Aug. 2023 - Present National Magnetic Resonance Research Center (UMRAM), Bilkent University Advisor: Assoc. Prof. Emine Ulku Saritas Engaged in research focused on advancing magnetic particle imaging (MPI) techniques for in vivo imaging using magnetic nanoparticles (MNPs). Designed and developed a magnetic particle spectrometer (MPS) device for characterizing the magnetization response of MNPs. Undergraduate Researcher, Aug. 2022 - Feb. 2023 Ultra Fast Laser Laboratory (UFOLAB), Bilkent University Advisor: Prof. Fatih Omer Ilday Conducted research on nonlinear feedback-driven laser material interactions for laser sintering techniques, utilizing lasers as power and heat sources to sinter powdered material based on 3D model-defined points, ultimately forming solid structures.
TEACHING EXPERIENCE	Teaching Assistant, Fall 2025 - Spring 2026 Bilkent University Course: Analog Electronics Assisted in conducting lab sessions, grading assignments, and holding office hours for undergraduate engineering students. Teaching Assistant, Fall 2025 - Spring 2026 Bilkent University Course: Electronic Circuit Design Supported course instruction, helped evaluate project designs, and provided mentorship on practical circuit implementation.

INTERNSHIPS

Candidate Engineer, Mar. 2025 - Aug. 2025

TÜBİTAK UZAY, Ankara, Turkey

Worked on satellite image processing and remote-sensing image analysis tasks, including data preparation, model evaluation, and image quality assessment.

Intern, Jun. 2024 - Sep. 2024

Institute of Biomedical Imaging (IBI), **Technical University of Hamburg**

Worked on simulation and design of Magnetic Particle Imaging subsystems. Conducted research to design and implement a 3D Helmholtz/Maxwell coil for magnetic sensor calibration.

Intern, Aug. 2023 - Sep. 2023

National Magnetic Resonance Research Center (UMRAM), **Bilkent University**

Conducted magnetic field simulations to design a magnetic particle spectrometer (MPS) setup. Performed 3D design of the Magnetic Particle Spectrometer setup.

Intern, Jun. 2023 - Jul. 2023

Helicopter Maintenance Factory, **Ministry of National Defense**

Worked on advancing a laser-based distance measurement system, involving a thorough exploration of theoretical advancements, including the development of innovative methodologies and algorithms to enhance precision and accuracy in distance measurements utilizing laser technology.

NOTABLE PROJECTS

MFH-MPS Hybrid System

Graduate Research (M.Sc.)

Combined magnetic fluid hyperthermia (MFH) and magnetic particle spectrometer (MPS) experimental platform, synchronizing AWG-driven heating with real-time temperature and MPS signal acquisition.

Magnetic Nanoparticle Imaging with a Single-Sided MPI

EEE 492 - Individual Research

Depth imaging using a single-sided Magnetic Particle Imaging to detect skin beneath nano-gels without selection fields.

3-Axis Helmholtz Coil

EEE 399 - Summer Internship

Design and implementation of 3D Helmholtz coil for Magnetic Particle Imaging head scanner sensor calibration.

Single-sided Magnetic Particle Spectrometer

EEE 491 - Individual Research

The development and application of a single-sided Magnetic Particle Spectrometer (MPS) device for detecting the response of magnetic nanoparticles (MNPs), for the ultimate goal of non-invasive tracking of the degradation of implanted MNPs/nano-gels.

High-Accuracy Initial Alignment System

Capstone Project - ROKETSAN

Developed a sensor-fusion alignment system for launch vehicles using low-precision IMUs and magnetometers, combining Gauss-Newton coarse alignment with a quaternion-based EKF and MATLAB/Arduino hardware validation.

Autonomous Aircraft

EEE 102 - Digital Circuit Design

Development and implementation of a drone capable of maintaining balance in the air using FPGA, with automatic takeoff and landing features.

COMPUTER SKILLS	Languages: MATLAB, Python, C++, Julia, VHDL, Assembly for Intel 8051, Spice Tools, Office Tools Applications: VSCode, COMSOL, Autodesk, Lightroom
LANGUAGES	Languages: English (IELTS Academic: 7), German (Beginner), Turkish (Native)